Problem 9.1 (MATLAB)
Signal models (AR, MA, ARMA) and parametric spectral estimation

In this problem is studied methods for parametric spectral estimation, in which the observed signal is assumed to be generated by linear filtering of zero mean white (Gaussian) noise. The signal models are

- **AR(p)**: Autoregressive model where the generating filter is an IIR filter with p poles and no zeroes
- **MA(q)**: Moving Average model where the generating filter is an FIR filter with q zeroes (and no poles)
- **ARMA(p,q)**: Autoregressive Moving Average model where the generating filter is an IIR filter with p poles and q zeroes

AR(p) model:

- Define a linear filter with p poles and no zeroes
- Generate an AR(p) signal by filtering white noise with the defined filter
- Estimate the signal PSD by using: Yule-Walker method (***pyulear***), covariance method (***pcov***), modified covariance method (***pmcov***) and Burg method (***pburg***)
- Use overlay plots to study random variations in the spectral estimation and plot also the mean estimate
- Study the influence of the signal length, N
- Study the influence of different assumed orders p_1, p_2, … (where the true order is p)
- What does the order selection criteria AIC and MDL suggest as optimal order ?
- Compare with non-parametric estimation, such as, e.g., the Welch method

MA(q) model:

- Define a linear filter with q zeroes and no poles
- Generate an MA(q) signal by filtering white noise with the defined filter
- Estimate the signal PSD by using the Blackman-Tukey method with a rectangular window of width 2q+1
- Use overlay plots to study random variations in the spectral estimation and plot also the mean estimate
- Study the influence of the signal length, N
- Study the influence of different assumed orders q_1, q_2, … (where the true order is q)

ARMA(p,q) model:

- Define a linear filter with p poles and q zeroes
- Generate an ARMA(p,q) signal by filtering white noise with the defined filter
- Estimate the signal PSD by using (***armax***)
- Use overlay plots to study random variations in the spectral estimation and plot also the mean estimate
- Study the influence of the signal length, N
- Study the influence of different assumed orders p_1, p_2, … and q_1, q_2, … (where the true orders are p and q)